C 23145 **(Pages : 3)** **Name.....**

Reg. No.....

SECOND SEMESTER M.Sc. DEGREE (SUPPLEMENTARY) EXAMINATION APRIL 2022

(CUCSS)

Mathematics

MT 2C 07—REAL ANALYSIS—II

(2016 to 2018 Admissions)

Time : Three Hours Maximum : 36 Weightage

Part A (Short Answer Questions)

Answer **all** *the questions.*
Each question carries a weightage of 1.

1. Show that constant functions are measurable.
2. Show that every countable set has measure zero.
3. Prove that outer measure is translation invariant.
4. Give an example where strict inequality occurs in Fatou's Lemma.
5. Show that if f is a non-negative measurable function, then $f = 0$ if and only if $\int f\, dx = 0$.
6. Show that if f and g are measurable, $|f| \leq |g|$ *a.e.*, and g is integrable, then f is integrable.
7. Show that BV $[a, b]$ is a vector space over the real numbers.
8. Show that the Lebesgue set of a function $f \in \mathrm{L}(a, b)$ contains any point at which f is continuous.
9. Show that Lebesgue measure m defined on M, the class of measurable sets of $\mathbb{R}$, is σ finite.
10. Show that $\mathrm{H}(\mathrm{R}) = \left\{\mathrm{E} : \mathrm{E} \subseteq \bigcup_{n=1}^{\infty} \mathrm{E}_n, \mathrm{E}_n \in \mathrm{R}\right\}$.
11. Give an example of a function which is continuous on $\mathbb{R}$ but not absolutely continuous.
12. Let f be a finite-valued monotone increasing function defined on (a, b), Show that $g(x) = f(x-)$ is left-continuous and monotone increasing on (a, b).

Turn over

13. Show that if $f \in L(a, b)$, its indefinite integral F is absolutely continuous on $[a, b]$.

14. Let μ be a complete measure on [[X, δ]] and $f : X \to \mathbb{R}$ is measurable, then μf^{-1} is a complete measure.

(14 × 1 = 14 weightage)

Part B

Answer any **seven** *questions.*
Each question carries a weightage of 2.

15. State and prove countably additive property.

16. Show that there exists uncountable sets of zero measure.

17. Prove that there exists a non-measurable set.

18. Show that $\int_0^1 \frac{\sin t}{e^t - x}\, dt = \sum_{n=1}^{\infty} \frac{x^{n-1}}{n^2 + 1}, -1 \le x \le 1.$

19. Prove that a function $f \in BV[a, b]$ if and only if f is the difference of two finite valued monotone increasing functions on $[a, b]$, where a and b are finite.

20. Let $[a, b]$ be a finite interval and let $f \in L(a, b)$ with indefinite integral F, then prove that $F' = f$ *a.e.* in $[a, b]$.

21. Let μ^* be the outer measure on $\mathcal{H}(\Re)$ defined by on $\Re$, then prove that δ^* contains $\delta(\Re)$, the σ ring generated by $\Re$.

22. Let μ be σ-finite measure and ν a σ-finite signed measure and let $\nu << \mu$; show that $\frac{d|\nu|}{d\mu} = \left|\frac{d\nu}{d\mu}\right| [\mu]$.

23. State and prove Jordan decomposition theorem.

24. Prove that every bounded linear functional F on C(I) can be written as $F = F^+ - F^-$, where F^+, F^- are positive linear functionals.

(7 × 2 = 14 weightage)

Part C

Answer any **two** *questions.*
Each question carries a weightage of 4.

25. Let c be any real number and let f and g be real-valued measurable functions defined on the same measurable set E. Then prove that $f + c$, cf, $f + g$, $f - g$ and fg are also measurable.

26. State and prove Lebesgue Dominated convergence theorem.

27. If μ is σ finite measure on a ring $\Re$, then prove that it has a unique extension to the σ ring $\delta(\Re)$.

28. State and prove Radon-Nikodym theorem.

($2 \times 4 = 8$ weightage)